

QRU • HEART HEALTH



INTERACTIVE LESSON

How the Human Heart Pumps Blood — Interactive Lesson

A Heart Health learning product

Foundational • General public

How the Human Heart Pumps Blood -
QRU PRESS™ Interactive Lesson

Edition 1

TREASURE STANDARD(TM) CERTIFIED

QRU Interactive Lesson - Heart Health

How the Human Heart Pumps Blood - Interactive Lesson

Manufactured by QRU Factory(TM). QRU simplifies the path to understanding the truth.

How the Human Heart Pumps Blood - Interactive Lesson

Suggested level: upper elementary to middle school

Estimated time: 15-25 minutes

Prerequisite: learners should know that blood carries oxygen and nutrients and that muscles can squeeze and relax.

The Question(TM)

How does the heart keep blood moving through my entire body every second of my life?

Before you begin, make a prediction:

- Which side of the heart sends blood to the lungs?
- Which side sends blood to the rest of the body?
- Why do you think blood needs to visit the lungs?

Interactive prediction check:

Write or select your answers now. Do not worry about being correct yet - you will revise them after the lesson.

| Prediction | My first idea | My revised answer |

|---|---|---|

| Which side sends blood to the lungs? | | |

| Which side sends blood to the body? | | |

| Why does blood visit the lungs? | | |

Simple Answer(TM)

Your heart is a strong muscular pump. It squeezes and relaxes again and again - about 100,000 times a day - to move blood through two connected loops:

- Right side of the heart ? lungs: sends oxygen-poor blood to the lungs to pick up oxygen and drop off carbon dioxide.
- Left side of the heart ? body: sends oxygen-rich blood to your brain, muscles, organs, and other body cells.

This is called double circulation because blood travels through two main loops:

1. Body ? right heart ? lungs
2. Lungs ? left heart ? body

Blood also carries nutrients and helps remove waste products such as carbon dioxide.

Quick check:

Circle or select the best answer.

1. The right side of the heart sends blood to the:

- A. body
- B. lungs
- C. brain only

2. The left side of the heart sends blood to the:

- A. body
- B. lungs
- C. stomach only

Feedback:

- If you chose lungs for 1, you are correct.
- If you chose body for 2, you are correct.
- If not, remember: Right = lungs. Left = body.

QRU Translation(TM)

Think of your heart as a four-room pump.

- The top rooms are the atria. They receive blood.
- The bottom rooms are the ventricles. They pump blood out.
- Valves are one-way doors that keep blood moving forward.
- An electrical signal tells the heart muscle when to squeeze.

A helpful comparison is a two-story house with four rooms, but remember: the analogy is not about the exact shape or height of the heart. It is about the job of each chamber:

- Atria = receiving rooms
- Ventricles = pumping rooms
- Valves = one-way doors

Important diagram tip: in most heart diagrams, the heart's right side appears on the viewer's left. Always follow the labels, not just the picture direction.

Key vocabulary scaffold

Start with the "must know" terms. Add the "stretch" terms when learners are ready.

| Must know term | Student-friendly meaning |

|---|---|

Atrium	Top chamber that receives blood
Ventricle	Bottom chamber that pumps blood out
Valve	One-way flap that keeps blood moving forward
Lungs	Organs where blood picks up oxygen and releases carbon dioxide
Body	The brain, muscles, organs, and cells that need oxygen
Stretch term	Student-friendly meaning
--- ---	
Vena cava	Large veins that bring oxygen-poor blood from the body to the right atrium
Pulmonary artery	Blood vessel that carries oxygen-poor blood from the right ventricle to the lungs
Pulmonary veins	Blood vessels that carry oxygen-rich blood from the lungs to the left atrium
Aorta	Large artery that carries oxygen-rich blood from the left ventricle to the body

Visual support to include

Use a clear labeled heart diagram with arrows showing direction of blood flow.

- Use blue arrows or a dashed pattern for oxygen-poor blood traveling to the lungs.
- Use red arrows or a solid pattern for oxygen-rich blood traveling to the body.
- Do not rely on color alone. Include:
 - labels
 - arrowheads
 - patterns
 - a legend/key
 - readable text
- Add labels for:
 - right atrium
 - right ventricle
 - left atrium
 - left ventricle
 - valves
 - lungs
 - body
 - vena cava
 - pulmonary artery
 - pulmonary veins
 - aorta

Accessibility note: include alt text such as:

"A labeled heart diagram showing oxygen-poor blood entering the right atrium from the body

through the vena cava, moving to the right ventricle and then to the lungs through the pulmonary artery. Oxygen-rich blood returns from the lungs through the pulmonary veins into the left atrium, moves to the left ventricle, and leaves through the aorta to the body."

Remember: diagram colors are learning tools. Real blood is not bright blue inside your body.

Interactive diagram task

Use the diagram to complete these actions:

1. Point to or click the right atrium.

This chamber receives oxygen-poor blood from the body.

2. Point to or click the right ventricle.

This chamber pumps blood to the lungs.

3. Point to or click the left atrium.

This chamber receives oxygen-rich blood from the lungs.

4. Point to or click the left ventricle.

This chamber pumps blood to the body.

5. Trace the arrows with your finger or cursor.

Say the full loop aloud:

Body ? right atrium ? right ventricle ? lungs ? left atrium ? left ventricle ? body

Feedback:

If your pathway crosses from the right side to the left side before visiting the lungs, check again.

Blood goes from the right heart to the lungs first, then returns to the left heart.

What's Happening Inside?(TM)

Follow the pathway of blood step by step:

1. From body to right atrium: Oxygen-poor blood returns from the body through large veins called the vena cava and enters the right atrium.

2. Right atrium to right ventricle: Blood moves through a one-way valve into the right ventricle.

3. Right ventricle to lungs: The right ventricle pumps blood through the pulmonary artery to the lungs.

4. In the lungs: Blood releases carbon dioxide and picks up oxygen.

5. Lungs to left atrium: Oxygen-rich blood returns through the pulmonary veins into the left atrium.

6. Left atrium to left ventricle: Blood moves through another one-way valve into the left ventricle.

7. Left ventricle to body: The left ventricle pumps oxygen-rich blood through the aorta to the

body.

8. Back to the heart again: Body cells use oxygen, and the blood becomes oxygen-poor again. It returns to the right atrium through the vena cava, and the cycle repeats.

Sequence challenge

Put these steps in the correct order:

- Blood travels through the pulmonary artery to the lungs.
- Oxygen-rich blood leaves the left ventricle through the aorta.
- Oxygen-poor blood enters the right atrium from the body.
- Blood moves from the left atrium to the left ventricle.
- Blood picks up oxygen in the lungs.
- Blood moves from the right atrium to the right ventricle.
- Oxygen-rich blood returns to the left atrium through the pulmonary veins.

Answer key:

1. Oxygen-poor blood enters the right atrium from the body.
2. Blood moves from the right atrium to the right ventricle.
3. Blood travels through the pulmonary artery to the lungs.
4. Blood picks up oxygen in the lungs.
5. Oxygen-rich blood returns to the left atrium through the pulmonary veins.
6. Blood moves from the left atrium to the left ventricle.
7. Oxygen-rich blood leaves the left ventricle through the aorta.

Valve mini-check

Valves act like one-way doors.

Choose the best answer:

1. Why are valves important?
 - A. They add oxygen to blood.
 - B. They keep blood moving in the correct direction.
 - C. They turn blood blue and red.
2. What might happen if there were no one-way valves?
 - A. Blood could move backward.
 - B. Blood would stop needing oxygen.
 - C. The lungs would disappear.

Answer key:

1. B. Valves keep blood moving in the correct direction.
2. A. Without one-way valves, blood could move backward.

Drag-and-drop or cut-and-sort activity

Move each item into the correct category.

| Goes with the right side of the heart | Goes with the left side of the heart |

|---|---|

|||

Cards to sort:

- oxygen-poor blood
- oxygen-rich blood
- pumps to lungs
- pumps to body
- right atrium
- right ventricle
- left atrium
- left ventricle
- pulmonary artery
- aorta

Answer key:

| Goes with the right side of the heart | Goes with the left side of the heart |

|---|---|

| oxygen-poor blood | oxygen-rich blood |

| pumps to lungs | pumps to body |

| right atrium | left atrium |

| right ventricle | left ventricle |

| pulmonary artery | aorta |

Check for understanding

Answer in one sentence each.

1. Why does blood need to go to the lungs?
2. Which chamber pumps blood to the lungs?
3. Which chamber pumps blood to the body?
4. Why is the left ventricle important?
5. What do valves do?

Answer key:

1. Blood goes to the lungs to pick up oxygen and release carbon dioxide.
2. The right ventricle pumps blood to the lungs.
3. The left ventricle pumps blood to the body.

4. The left ventricle is important because it pumps oxygen-rich blood out through the aorta to the body.
5. Valves keep blood moving forward and help prevent backward flow.

Explain it like a scientist

Use these words to explain the full pathway:

right atrium, right ventricle, lungs, left atrium, left ventricle, body

Sentence frame:

Oxygen-poor blood comes from the _____ and enters the _____. Then it moves to the _____, which pumps it to the _____. After picking up oxygen, blood returns to the _____, moves to the _____, and is pumped to the _____.

Answer key:

Oxygen-poor blood comes from the body and enters the right atrium. Then it moves to the right ventricle, which pumps it to the lungs. After picking up oxygen, blood returns to the left atrium, moves to the left ventricle, and is pumped to the body.

Differentiation

For upper elementary learners:

- Focus on the main loop:
Body ? right heart ? lungs ? left heart ? body
- Use the terms atrium, ventricle, valve, lungs, and body first.
- Add vessel names after learners understand the pathway.

For middle school learners:

- Add the vessel names: vena cava, pulmonary artery, pulmonary veins, and aorta.
- Ask why the pulmonary artery is unusual: it carries oxygen-poor blood away from the heart.
- Ask why the pulmonary veins are unusual: they carry oxygen-rich blood toward the heart.

Exit ticket

Complete these before finishing:

1. The right side of the heart sends blood to the _____.
2. The left side of the heart sends blood to the _____.
3. Blood picks up oxygen in the _____.
4. The aorta carries blood from the _____ to the _____.
5. Write the full pathway in arrows:
? ? ? ?

Exit ticket answer key:

1. lungs
2. body
3. lungs
4. left ventricle; body
5. Body ? right heart ? lungs ? left heart ? body

Continue your understanding at QRU:

